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Elastic perfect plastic rock mass analysis and equivalent *GSI* determination considering strain-softening behavior and three-dimensional strength

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A R T I C L E I N F O

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Equivalent Geological Strength Index (GSI_{eq})

A B S T R A C T

The elastic perfect plastic model (EPP) fails to accurately capture the strain-softening mechanical behaviors of rock masses, while the post-peak constitutive model of elastic strain softening (ESS) has great application limitations in practical engineering for complex numerical algorithms. In this paper, an EPP constitutive approximation of rock mass considering three-dimensional strength and strain-softening characteristics is proposed. The forward analysis method of the equivalent Geological Strength Index (GSI_{eq}) is given considering the impact of in-situ stress, rock mass properties and engineering excavation. Extensive case studies demonstrate that the radial stress and displacement obtained by the proposed method agree well with those of the ESS model, with only small errors in the Ground Response Curve. The rock mass elastic modulus E and the initial stress level p_0 significantly affect GSI_{eq}, while the support pressure p_s contributes little to that. As $R_{p,EPP}/R_0 > 3.0$, usually $GSI < 25$, the residual strength parameter GSI_r can be directly used as the calculation parameter of the EPP model, while for a highly qualified rock mass, such as $GSI > 75$, the peak strength parameter GSI_p can be directly used as an approximate value of GSI_{eq}. Furthermore, with the increase of the in-situ stress level, GSI_{eq} gradually decreases to GSI_r, and the difference of plastic stress distribution law based on ESS and EPP models gradually diminishes. Moreover, the proportion of plastic deformation to total deformation increases nonlinearly with buried depth. The proposed method offers a new calculation idea for the simplified analysis of strain-softening rock masses.

1. Introduction

The stability analysis of surrounding rock and the pre-design of support for tunnels after excavation disturbance require careful consideration of stress, displacement, and plastic zone distribution. The complex mechanical behavior of rock mass is the key to the simulation of tunnel excavation. Better consideration of the mechanical behavior of rock mass in the analysis is helpful for the economical design of tunnel structures. Hoek and Brown (1997) recommended using the elastic perfect plastic (EPP), elastic strain softening (ESS), and elastic brittle plastic (EBP) models to describe the post-peak mechanical behavior of rock mass from low to high quality, respectively. In practical engineering, most rock masses exhibit strain-softening characteristics. However, due to the uncertainty of input parameters and softening law assumptions, complexity of calculation formulas, difficulty of numerical implementation, along with tedious calculation time consumption, the application and promotion of the ESS model is greatly limited in practical engineering (Cui et al., 2019). In contrast, the EPP model is widely used for theoretical analysis of geotechnical engineering problems and finite element calculations of practical problems (Cai et al., 2022b; Li et al., 2018; Li et al., 2022) for its advantages of simpler model, more mature analysis technology and faster calculation speed, despite its deficiencies in capturing real strain-softening behavior of rock mass (Cui et al., 2017). To the best of knowledge, the research on the analysis method is to find a balance between the two contradictory points, namely the fineness of the result and the speed of calculation, according to the actual engineering needs.

Previous studies have developed various calculation methods for surrounding rock response using different constitutive models (González-Cao et al., 2013; Guan et al., 2007; Kabwe, 2020; Reed, 1988). However, current elastoplastic analysis methods primarily rely on two-dimensional strength theory, ignoring the influence of

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intermediate principal stress and focusing on the shallow strata under a low-stress state. In contrast, for deep rock mass engineering under high-stress conditions, accurate analysis of surrounding rock stability demands higher requirements for three-dimensional strength criteria and constitutive relations (Cai et al., 2022a; Cai et al., 2023; Zhao et al., 2023a). Moreover, the Convergence-Confinement Method (CCM) is commonly used in engineering to reveal the deformation and failure mechanism of surrounding rock. Sun et al. (2022) considered the installation sequence and length of anchor bolts and cables, and analyzed the whole-process interaction between the anchor cable combination and the surrounding rock along the tunnel axis using Longitudinal Displacement Profile (LDP) and Ground Response Curve (GRC). Similarly, Chu et al. (Chu et al., 2019; Chu et al., 2021) well revealed the time-dependent interaction between the surrounding rock and support system considering the coupling effect of stress release and rock creep through a novel Burgers viscoelastic analytical solution. However, how to consider the strain-softening characteristics of rock mass is a major difficulty in the application of the CCM method (Alejano et al., 2010). Most ESS constitutive models are complex (Huang et al., 2018; Li et al., 2012; Pourhosseini & Shabanimashcool, 2014), and the parameter selection and softening law assumptions of simplified models are also different (Ghorbani & Hasanzadehshooiili, 2019; Wang & Qian, 2018; Zareifard, 2020). Consequently, plasticity analysis involving strain-softening properties often necessitates the use of numerical methods to solve partial differential equations (Carranza-Torres, 2004; Park, 2014; Sharan, 2008; Zhang et al., 2021), or finite difference method (Lee & Pietruszczak, 2008; Zhang et al., 2012). When a three-dimensional nonlinear strength criterion with a more complex expression is introduced, the response of the plastic zone with the ESS model is more challenging (Xu et al., 2021). The application of the ESS model in finite element numerical calculations is more complicated. Softening materials have defects in the constitutive integration process based on classical plasticity theory, necessitating the development of specific constitutive integration algorithms (Jin et al., 2020; Ren et al., 2023; Wang & Jiang, 2015; Wang et al., 2011; Zhao et al., 2023b). Additionally, for non-smooth yield criteria in the principal stress space, such as the Hoek–Brown and GZZ criteria, it is crucial to smooth the yield surface to eliminate the singularity and ensure the convexity of the yield surface for the accuracy and convergence of numerical calculations (Clausen & Damkilde, 2008; Dai et al., 2018).

Due to the complexity of directly solving the ESS problem, some studies have extended the solution of the EPP problem to obtain that of ESS problem. Alejano et al. (2012) gave the difference formula of the plastic zone radius between ESS and EPP rock mass tunnel by statistical analysis, and proposed an approximation equation for the plastic radius of the ESS rock mass tunnel. Therefore, the LDP calculation method proposed by Vlachopoulos and Diederichs (2009), suitable for the EPP model considering the plastic radius, is extended and applied to the ESS rock mass. For the numerical analysis of the Hoek–Brown rock mass considering the strain-softening characteristics, the Mohr–Coulomb criterion is often used instead because it is more convenient for expressing these characteristics, or an EPP Hoek–Brown model is used by selecting appropriate parameters (Sørensen et al., 2015). However, due to the uncertainty in the field and laboratory tests, the value of rock mass parameters is an essential factor affecting the analysis results (Song et al., 2016; Tiwari et al., 2017). To address this issue, Hoek et al. (2002) proposed a value method for determining the equivalent Mohr–Coulomb criterion parameters c' and φ' by fitting the relative positions of the Hoek–Brown and Mohr–Coulomb criteria in the principal stress space. However, the parameter selection of the equivalent EPP Hoek–Brown model has not been studied before.

It is worth noting that there is no distinct boundary between ESS and EPP models, and thus they can transform into each other under certain conditions. The ESS model can transform into the EPP model when the strength drop modulus of the ESS model approaches to zero, or when the residual strength is equal to the peak strength. Several studies have conducted similar analyses (Song & Rodriguez-Dono, 2021; Zhang et al., 2021). Moreover, laboratory tests have demonstrated that the rock in the post-peak state exhibits strain-softening behavior under low confining pressure, and gradually evolves to elastic perfect plastic as the confining pressure increases (Feng et al., 2021; Xie et al., 2021). Wu et al. (2018) developed an ESS model considering the positive correlation between residual strength and confining pressure to describe the changes in mechanical response characteristics of rock masses as they transition from ESS to EPP with increasing confining pressure. Actually, strain-softening behavior refers to the gradual loss of strength, or weakening, of the rock mass (Cai et al., 2007). On the other hand, this behavior is mostly manifested as a reduction in the bearing capacity of the rock mass after cracks are localized and penetrated inside the rock mass, rather than the inherent material properties of the rock mass. In our previous work (Su et al., 2022), we compared the calculation time consumption of the ESS and EPP models, and found a significant difference of thousands of times in the calculation efficiency of plastic stress. Therefore, by reducing the peak Geological Strength Index (*GSI*) of the rock mass in advance to account for strength weakening and substituting the weakened *GSI* into the EPP model for calculation, one can significantly improve calculation efficiency while obtaining satisfactory calculation accuracy, making it a reasonable treatment method.

Given the above problems, an approximate idea of an EPP constitutive model of rock mass considering the three-dimensional strength and strain-softening characteristics is proposed first in this paper. Then, considering the influence of in-situ stress level, rock mass characteristics and engineering excavation, the forward analysis value method of equivalent *GSI* (GSI_{eq}) is given through extensive case analyses. Finally, the results correctness and calculation efficiency of the idea and parameter value method in the stability analysis of surrounding rock are verified.

2. Problem description

2.1. Calculation model of tunnel excavation

The response of surrounding rock following the excavation of a deep-buried circular tunnel is a classic problem in rock mechanics that can be simplified as a “thick-walled cylinder” problem. It is assumed that the initial geostress condition of the homogeneous isotropic surrounding rock is the hydrostatic pressure p_0 in the excavation plane and the axial

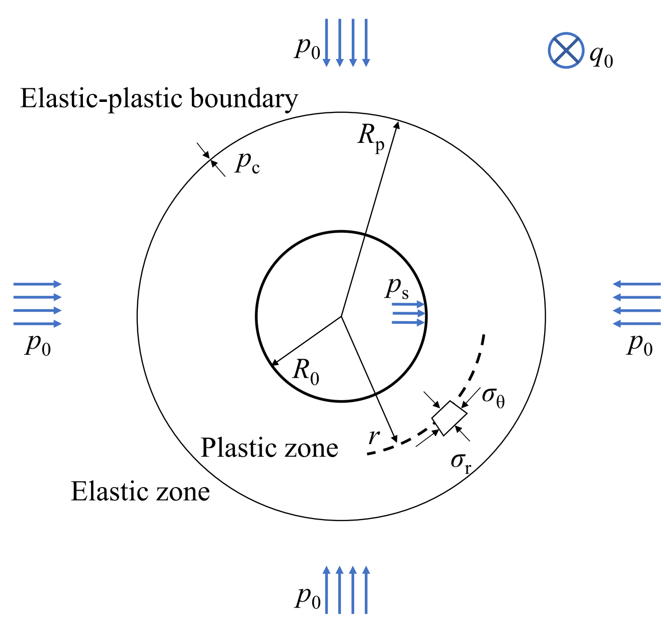

Fig. 1. Tunnel excavation model.

stress q_0 outside the plane, as shown in Fig. 1. After excavation, the inner diameter R_0 of the tunnel is subjected to the support pressure p_s, while the stress at the plastic zone radius R_p is p_c. p_c is the critical stress at which the plastic zone exists or not. When $p_s < p_c$ the supporting stress is insufficient, the plastic zone will appear in the surrounding rock.

This paper uses the three-dimensional Hoek–Brown strength criterion, namely the Generalized Zhang and Zhu (GZZ) strength criterion (Zhang, 2008; Zhang & Zhu, 2007), to determine whether the surrounding rock enters plasticity. The modified GZZ criterion (Cai et al., 2021) in terms of stress invariants is expressed as follows:

$$F = \frac{1}{\sigma_c^{(1/a-1)}}\left(\sqrt{3J_2}\right)^{1/a} + \left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\frac{I_1}{3} - s\sigma_c = 0 \tag{1}$$

where σ_c represents the uniaxial compressive strength of the rock; θ_σ is the Lode angle; I_1 is the first invariant of the stress tensor; and J_2 is the second invariant of the stress deviation. Additionally, m_b, s, and a are empirical parameters reflecting the characteristics of rock mass, and can be obtained through the conversion of rock mass *GSI*, m_i, and D,

$$\begin{cases} m_b = m_i \exp[(GSI-100)/(28-14D)] \\ s = \exp[(GSI-100)/(9-3D)] \\ a = 1/2 + [\exp(-GSI/15) - \exp(-20/3)]/6 \end{cases} \tag{2}$$

where *GSI* is the Geological Strength Index, which reflects the joint density and joint strength behavior of the rock mass; m_i is the laboratory rock parameter; and D is the disturbance parameter considering blasting and stress release, which can be determined directly according to the recommendation of Hoek et al. (2002).

For the mechanical behavior of surrounding rock in the plastic zone, Hoek and Brown (1997) suggest, based on significant rock mass engineering experience, that the EPP model is suitable for weak surrounding rock with $GSI < 25$, the EBP model is appropriate for relatively intact rock with $GSI > 75$, while the ESS model is recommended for the medium-quality rocks with $25 < GSI < 75$ (Alejano et al., 2010; Ghorbani & Hasanzadehshooiili, 2017), as shown in Fig. 2.

The post-peak strain-softening behavior of the rock mass is characterized by the Geological Strength Index (*GSI*). The rock strength parameters weaken with the increase of plastic strain (Hou & Cai, 2023; Park et al., 2008; Zareifard, 2020). It is assumed that the *GSI* decreases linearly with the increase of the plastic deviatoric strain γ^p in the softening stage, and the residual value remains constant after reaching the critical value,

$$GSI(\gamma^p) = \begin{cases} GSI_p - (GSI_p - GSI_r)\cdot\gamma^p/\gamma^{p^*} & 0 < \gamma^p < \gamma^{p^*} \\ GSI_r & \gamma^p \geqslant \gamma^{p^*} \end{cases} \tag{3}$$

where $\gamma^p = \varepsilon_\theta^p - \varepsilon_r^p$, γ^{p^*} is the critical plastic deviatoric strain, GSI_p and GSI_r are the peak and residual values of the *GSI*, respectively. The instability problems such as bifurcation and strain localization in strain-softening regime are not discussed here. The residual value of *GSI* for a given rock mass can be determined using an empirical formula proposed by Cai et al. (2007), which has validated by in-situ test data of large-scale cavern construction sites and is expressed as follows:

$$GSI_r = GSI_p \exp(-0.0134 GSI_p) \tag{4}$$

2.2. Comparison of surrounding rock response of different post-peak models

Different post-peak mechanical behaviors of rock mass correspond to different stress and displacement distributions. To investigate the differences between the widely used EPP model and the more adaptable ESS model, the solution method in Part 3 is used to solve the excavation response of a strain-softening rock mass. The rock mass parameters used are as follows: $GSI_0 = 50$, $GSI_r = 25.6$, $m_i = 10$, $D = 0$, $\sigma_c = 40$ MPa, $E = 5$ GPa, $\upsilon = 0.25$, $\gamma^{p^*} = 0.02$, $\psi = 10^\circ$, excavation radius $R_0 = 8$ m, $p_0 = q_0 = 30$ MPa, and $p_s = 1$ MPa. The calculation results obtained from the ESS and the EPP models are compared, with the latter calculated using GSI_p and GSI_r. The comparison of the calculation results is presented in Fig. 3.

Compared to the ESS model, the EPP model with GSI_p overestimates the rock mass quality in the plastic zone, resulting in larger stress in the plastic zone, smaller displacement of surrounding rock, and a smaller range of plastic zone. Conversely, the EPP model directly using GSI_r ignores the rock mass quality in the softening zone, and calculates with the lowest rock mass quality, resulting in smaller stress, larger displacement, and wider distribution of the plastic zone. When the EPP model is used instead of the ESS model for engineering calculations, directly using either the peak value or residual value will lead to significant errors. In fact, the shape of the displacement curve of the EPP model is not significantly different from that of the ESS model in Fig. 3, and the stress curve is not much different. Therefore, an optimal *GSI* value can be found between the GSI_p and GSI_r, so that the results calculated by the EPP model are similar to those calculated by the ESS model, as shown by the green curve in Fig. 3.

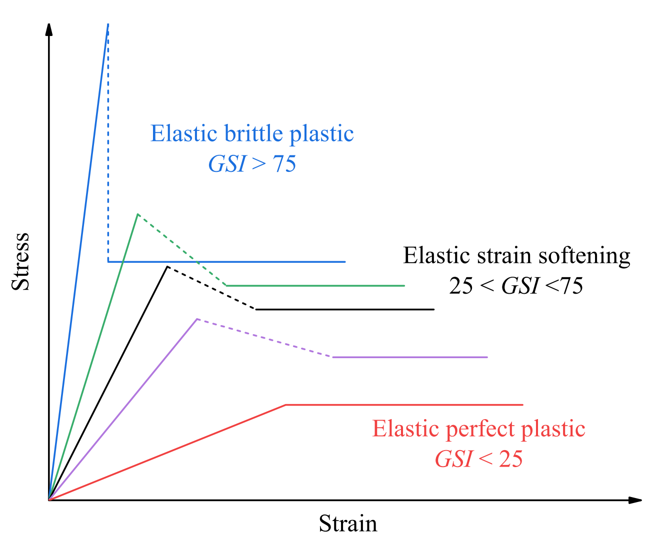

Fig. 2. Different post-peak rock mass behaviors. Based on Hoek and Brown (1997).

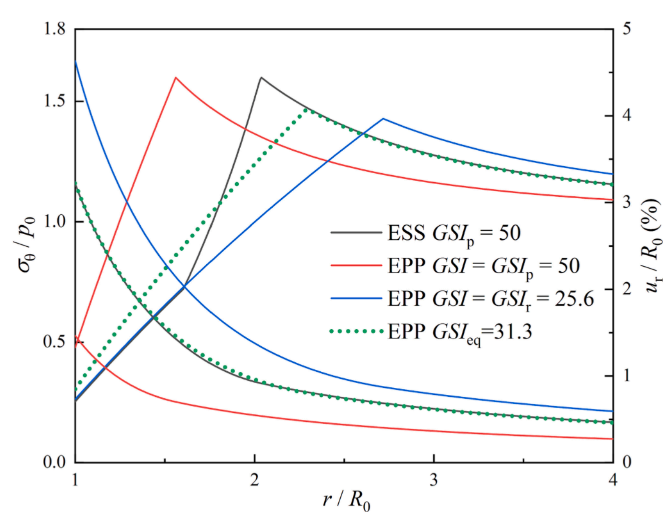

Fig. 3. Results comparison of ESS and EPP models with different calculation parameters.

2.3. Equivalent GSI

To determine the GSI_{eq} value of the EPP model that produces results similar to those of the ESS model, an objective function consisting of the displacement and the stress terms is used to represent the difference between the two calculation models, which is expressed as follows:

$$T(GSI) = \frac{1}{\sqrt{M}}\sqrt{\sum_{i=1}^{M}\left(\frac{u_{EPP}^{i} - u_{ESS}^{i}}{u_{ESS}^{i}}\right)^2} + \frac{\alpha}{\sqrt{N}}\sqrt{\sum_{j=1}^{N}\left(\frac{\sigma_{EPP}^{j} - \sigma_{ESS}^{j}}{\sigma_{ESS}^{j}}\right)^2} \tag{5}$$

where M, $N = 1, 2, \ldots$; u and σ represent the eigenvalues obtained by the two models at various locations, such as the tunnel wall, the elastic–-plastic boundary, and the boundary of flow and softening zones; and α is an adjustment coefficient used to ensure consistency between the values of the two items. The difference between the two models varies with different GSI values. When T reaches its minimum, the corresponding GSI is defined as the equivalent Geological Strength Index (GSI_{eq}). In this paper, $M = N = 1$, $\alpha = 1$, the displacement of the tunnel wall, u_{R0}, is taken as the displacement eigenvalue, and the radial stress at the tunnel wall, $\sigma_{r,R0}$, is taken as the stress eigenvalue, for the calculation of $T(GSI)$.

The relative positional relationship of GSI_{eq} between GSI_p and GSI_r can be represented by β, where $\beta = (GSI_{eq} - GSI_r)/(GSI_p - GSI_r)$, and $0 < \beta \leq 1$ since $GSI_r < GSI_{eq} < GSI_p$. When $\beta = 1$, $GSI_{eq} = GSI_p$, there is no plastic zone, and the plastic zone radius of the ESS model $R_{p,ESS}$ is equal to R_0. When the plastic zone is widely distributed, usually with $R_{p,ESS} > 3R_0$, the GSI_{eq} value is closer to GSI_r, and the value of β is smaller and close to 0. The plastic zone radius $R_{p,ESS}$ comprehensively reflects the quality of the surrounding rock and the geomechanical environment, and has a good correspondence with β. Therefore, it is reasonable and reliable to use $R_{p,ESS}$ to calculate β. However, $R_{p,ESS}$ requires the calculation of the ESS model, which is contrary to the goal of this paper to avoid the complicated solution of the ESS model. The $R_{p,EPP}$ calculated by the EPP model and GSI_p has a good correspondence with $R_{p,ESS}$. In this paper, the easy-to-calculate $R_{p,EPP}$ is used instead of $R_{p,ESS}$ to characterize the variation of β (ie, GSI_{eq}).

To analyze the relationship between β and $R_{p,EPP}$, the values of GSI_{eq} in the range of $GSI_p = 20 - 80$ are calculated by the method described above. To ensure consistency across different GSI_p within the same calculation set, we use the same GSI softening rate, denoted as $\chi = (GSI_p - GSI_r)/\gamma^{p^*}$. It is worth noting that the value of γ^{p^*} varies with GSI_p in one set of calculation cases. The resulting scatter points of the four sets of parameter calculations are plotted in Fig. 4.

An approximate exponential function relationship between β and $R_{p,EPP}/R_0$ is evident in Fig. 4. The following formula is used to fit the calculation results,

$$\beta = \frac{(GSI_{eq} - GSI_r)}{(GSI_p - GSI_r)} = \exp\left(k\cdot\frac{R_{p,EPP}}{R_0}\right) \Big/ \exp(k) \tag{6}$$

where k is an undetermined coefficient, and $R_{p,EPP}$ is the plastic zone radius calculated by the EPP model with GSI_p, which can reflect the quality and geological conditions of the surrounding rock. When $R_{p,EPP} = R_0$, the surrounding rock is in an elastic state, $GSI_{eq} = GSI_p$ calculated by Eq. (6), and the peak strength can be directly used for calculation. This implies the tunnel has good stability and may not require a support system. If $R_{p,EPP}/R_0$ exceeds 3.0, the plastic zone is relatively large, in which a small range of softening zone can be ignored, and GSI_r can be directly used for calculation, resulting in conservative results with a small error. Eq. (6) establishes the functional relationship between $R_{p,EPP}$ and β, and GSI_{eq} can be obtained by solving $R_{p,EPP}$ as shown in Eq. (7). The value of k will be discussed in Part 4.

$$GSI_{eq} = GSI_r + (GSI_p - GSI_r)\exp\left[k\cdot\left(\frac{R_{p,EPP}}{R_0} - 1\right)\right] \tag{7}$$

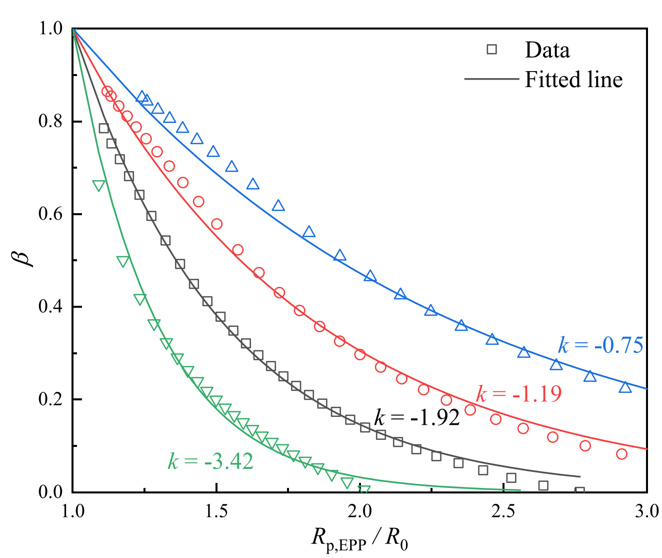

Fig. 4. Fitting results.

3. Elasto-plastic response calculation method

3.1. Solution of elastic zone

The radial stress σ_r, tangential stress σ_θ and axial stress σ_z in the elastic zone can be expressed as:

$$\begin{cases} \sigma_r = p - (p - p_s)R_0^2/r^2 \\ \sigma_\theta = p + (p - p_s)R_0^2/r^2 \\ \sigma_z = q \end{cases} \tag{8}$$

In the excavation plane, the radial strain ε_r, tangential strain ε_θ and radial displacement u_r of surrounding rock can be expressed as:

$$\begin{cases} \varepsilon_r = (-p + p_s)R_0^2/(2Gr^2) \\ \varepsilon_\theta = (p - p_s)R_0^2/(2Gr^2) \\ u_r = (p - p_s)R_0^2/(2Gr) \end{cases} \tag{9}$$

where $G = E/(1 + \nu)/2$ is the shear modulus, and ν is Poisson's ratio. When the support stress is less than the critical stress of the plastic zone existing, $p_s < p_c$, the plastic zone begins to appear in the rock mass around the tunnel. The solution of the elastic zone response can be obtained when the plastic zone exists by replacing R_0 and p_s in Eqs. (8) and (9) with R_p and p_c, respectively.

3.2. Stress solution of plastic zone

The expression for the three-dimensional nonlinear GZZ strength criterion is a transcendental equation, making it impossible to separate the principal stresses and obtain a strict analytical solution. Therefore, the response of plastic zone is solved using the finite difference method (Su et al., 2022) in this study.

In the finite difference method, the surrounding rock in the plastic zone is divided into n rings, where $i = 0$ at the elastic–plastic boundary, and $i = n$ at the tunnel wall, as shown in Fig. 5. The radial stress difference $\Delta\sigma_r = \sigma_{r(i)} - \sigma_{r(i-1)} = -(p_c - p_s)/n$ between each ring is the same, but the thickness differs. The inner diameter of each ring in the plastic zone is normalized, and the relative radius ρ is taken as:

$$\rho_{(i)} = r_{(i)}/R_p \tag{10}$$

Similarly, the relative radius ρ is $\rho_{(0)} = 1$ at the elastic–plastic boundary, and $\rho_{(n)} = R_0/R_p$ at the tunnel wall.

The stress component is calculated from the elastic–plastic boundary

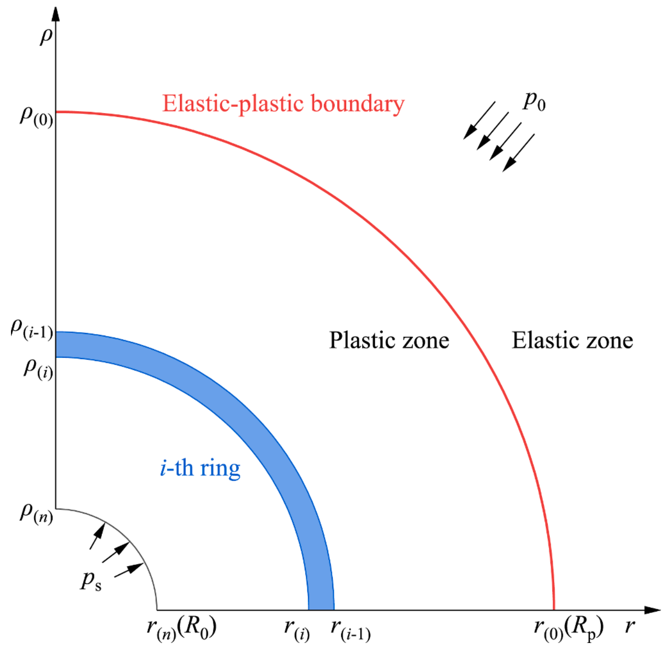

Fig. 5. Plastic zone ring division.

to the tunnel wall. When $i = 0$, $r_{(0)} = R_p$, it can be obtained from Eqs. (8) and (9) that:

$$\begin{Bmatrix} \sigma_{r(0)} \\ \sigma_{\theta(0)} \\ \sigma_{z(0)} \\ \varepsilon_{r(0)} \\ \varepsilon_{\theta(0)} \end{Bmatrix} = \begin{Bmatrix} p_c \\ 2p - p_c \\ q \\ (p_c - p)/(2G) \\ (p - p_c)/(2G) \end{Bmatrix} \tag{11}$$

The stress at the elastic–plastic boundary should also satisfy the consistency condition of the plastic zone. The radial stress p_c at the elastic–plastic boundary can be obtained by substituting Eq. (11) into Eq. (1). Eq. (11) has provided the necessary initial boundary condition, as required by the finite difference method.

In the stress calculation process at each ring, the stress components ($\sigma_{r(i)}$, $\sigma_{r(i)}$, $\sigma_{r(i)}$) on the inner side ($\rho_{(i)}$) of the i-th ring needs to be solved, while the stress components $\boldsymbol{\sigma}_{(i\text{-}1)}$ on the outer side ($\rho_{(i\text{-}1)}$) are known. Given that $\sigma_{r(i)} = \sigma_{r(i\text{-}1)} + \Delta\sigma_r$ is known, only two unknowns, $\sigma_{\theta(i)}$ and $\sigma_{z(i)}$, remain to be solved. Two equations are needed for this purpose, usually the plastic zone consistency Eq. (1) and a supplementary equation for out-of-plane stress. For the plastic zone consistency equation $F(\boldsymbol{\sigma}_{(i)}) = 0$, when $F(\boldsymbol{\sigma}_{(i\text{-}1)}) = 0$ is known, it can be equivalently replaced by:

$$\mathrm{d}F = F\left(\boldsymbol{\sigma}_{(i)}\right) - F\left(\boldsymbol{\sigma}_{(i\text{ - }1)}\right) = 0 \tag{12}$$

For the plastic flow section of the EPP model (Lubliner, 2008), Eq. (12) can be written as:

$$\begin{aligned} dF &= \frac{\partial F}{\partial \sigma} d\sigma \\ &= \frac{\partial F}{\partial \sigma_r} d\sigma_r + \frac{\partial F}{\partial \sigma_\theta} d\sigma_\theta + \frac{\partial F}{\partial \sigma_z} d\sigma_z \\ &= \frac{\partial F}{\partial \sigma_{r(i-1)}}\left(\sigma_{r(i)} - \sigma_{r(i-1)}\right) + \frac{\partial F}{\partial \sigma_{\theta(i-1)}}\left(\sigma_{\theta(i)} - \sigma_{\theta(i-1)}\right) + \frac{\partial F}{\partial \sigma_{z(i-1)}}\left(\sigma_{z(i)} - \sigma_{z(i-1)}\right) \\ &= 0 \end{aligned} \tag{13}$$

The differential relational expression of the stress component between the inner and outer radii is established at $\rho_{(i-1)}$ in Eq. (13), which yields the linear expression of $\sigma_{\theta(i)}$ and $\sigma_{z(i)}$. The partial derivative of the GZZ strength criterion Eq. (1) with respect to each principal stress is:

$$\frac{\partial F}{\partial \sigma} = m_b\left[\frac{1}{6} + \frac{\xi(J_2, \theta_\sigma)}{2\sqrt{J_2}} S - \frac{3}{4J_2} S \cdot S\right] \tag{14}$$

where $\mathbf{S}$ is the stress deviator tensor, and $\xi(J_2, \theta_\sigma) = \sqrt{3}\left[\left(\sqrt{3J_2}\right)^{1/a-1} / a m_b \sigma_c^{(1/a-1)} - \sin 3\theta_\sigma / 3 + 1/2\right]$.

To simplify the calculation, the supplementary equation for out-of-plane stress is given as:

$$\sigma_{z(i)} = \left(\sigma_{\theta(i)} + \sigma_{r(i)}\right)/2 \tag{15}$$

The explicit expressions of $\sigma_{\theta(i)}$ and $\sigma_{z(i)}$ can be quickly obtained by combining the two linear equations, namely Eqs. (13) and (15):

$$\begin{cases} \sigma_{\theta(i)} = \dfrac{L_1 c_{z2} - L_2 c_{z1}}{c_{\theta 1} c_{z2} - c_{\theta 2} c_{z1}} \\ \sigma_{z(i)} = \dfrac{L_1 c_{\theta 2} - L_2 c_{\theta 1}}{c_{z1} c_{\theta 2} - c_{z2} c_{\theta 1}} \end{cases} \tag{16}$$

where:

$$\begin{cases} c_{\theta 1} = \dfrac{\partial F}{\partial \sigma_{\theta(i-1)}}, c_{z1} = \dfrac{\partial F}{\partial \sigma_{z(i-1)}}, c_{\theta 2} = \dfrac{1}{2}, c_{z2} = -1 \\ L_1 = \dfrac{\partial F}{\partial \sigma_{\theta(i-1)}} \sigma_{\theta(i-1)} + \dfrac{\partial F}{\partial \sigma_{z(i-1)}} \sigma_{z(i-1)} - \dfrac{\partial F}{\partial \sigma_{r(i-1)}} \Delta\sigma_r \\ L_2 = -\dfrac{1}{2}\sigma_{r(i)} \end{cases} \tag{17}$$

In the ESS model, the strength parameters are not constant, which means that Eq. (13) is not applicable. Therefore, the numerical iterative method must be used to solve the yield criterion Eq. (1) to determine the plastic stress at the inner radius of each ring (Xia et al., 2018). Although the EPP model can be solved directly like the ESS model, the proposed method replaces the yield criterion Eq. (1) with a linear equation Eq. (13). This replacement eliminates the need to iteratively solve nonlinear equations and greatly enhances computational efficiency.

3.3. Strain solution of plastic zone

The stress components of the plastic zone should satisfy the equilibrium equation, expressed as:

$$\frac{d\sigma_r}{dr} - \frac{\sigma_\theta - \sigma_r}{r} = 0 \tag{18}$$

After normalizing the radius, Eq. (18) can be expressed in a difference form as:

$$\frac{\Delta\sigma_r}{\rho_{(i)} - \rho_{(i-1)}} - \frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{\rho_{(i)}} = 0 \tag{19}$$

As the stress solution has been obtained, the inner side $\rho_{(i)}$ of the i-th ring can be calculated using the known outer side $\rho_{(i\text{-}1)}$.

The strain component $\varepsilon_{(i)}$ can be decomposed into elastic and plastic strains as:

$$\begin{Bmatrix} \varepsilon_{r(i)} \\ \varepsilon_{\theta(i)} \end{Bmatrix} = \begin{Bmatrix} \varepsilon^e_{r(i)} \\ \varepsilon^e_{\theta(i)} \end{Bmatrix} + \begin{Bmatrix} \varepsilon^p_{r(i)} \\ \varepsilon^p_{\theta(i)} \end{Bmatrix} \tag{20}$$

The elastic strain can be determined using Hooke's law, while the plastic strain can be calculated using the following method.

The plastic zone strain should satisfy the compatibility equation, shown as:

$$\frac{d\varepsilon_\theta}{dr} + \frac{\varepsilon_\theta - \varepsilon_r}{r} = 0 \tag{21}$$

The strain component is separated into elastic and plastic parts, and the Eq. (21) is used in the inner and outer sides of the i-th ring to get:

$$\frac{\left(\varepsilon^{p}_{\theta(i)}-\varepsilon^{p}_{\theta(i-1)}\right)+\left(\varepsilon^{e}_{\theta(i)}-\varepsilon^{e}_{\theta(i-1)}\right)}{r_{(i)}-r_{(i-1)}}+\frac{\left(\varepsilon^{p}_{\theta(i)}-\varepsilon^{p}_{r(i)}\right)+\left(\varepsilon^{e}_{\theta(i)}-\varepsilon^{e}_{r(i)}\right)}{r_{(i)}}=0 \tag{22}$$

Assuming a non-associated flow rule, the plastic radial and tangential strains are related as follows:

$$d\varepsilon^{p}_{r}+\lambda d\varepsilon^{p}_{\theta}=0 \tag{23}$$

where λ is the dilatation coefficient, which is related to the dilation angle as follows:

$$\lambda=\frac{1+\sin\psi}{1-\sin\psi} \tag{24}$$

When the dilatation angle ψ equals 0, the dilatation coefficient λ is 1, indicating that the plastic volume strain is not taken into consideration.

The difference form of Eq. (23) is:

$$\varepsilon^{p}_{r(i)}-\varepsilon^{p}_{r(i-1)}+\lambda\left(\varepsilon^{p}_{\theta(i)}-\varepsilon^{p}_{\theta(i-1)}\right)=0 \tag{25}$$

The expression of plastic strain for the inner side of each ring can be obtained by combining the Eqs. (22) and (25):

$$\begin{cases}\varepsilon^{p}_{\theta(i)}=\dfrac{\varepsilon^{p}_{\theta(i-1)}+\eta_{(i)}\left(\lambda\varepsilon^{p}_{\theta(i-1)}+\varepsilon^{p}_{r(i-1)}\right)-\left(\varepsilon^{e}_{\theta(i)}-\varepsilon^{e}_{\theta(i-1)}\right)-\eta_{(i)}\left(\varepsilon^{e}_{\theta(i)}-\varepsilon^{e}_{r(i)}\right)}{1+(1+\lambda)\eta_{(i)}}\\ \varepsilon^{p}_{r(i)}=\varepsilon^{p}_{r(i-1)}-\lambda\left(\varepsilon^{p}_{\theta(i)}-\varepsilon^{p}_{\theta(i-1)}\right)\end{cases} \tag{26}$$

where $\eta_{(i)} = 1-r_{(i-1)}/r_{(i)}$.

After n calculations, $\rho_{(n)}$ is obtained. Substituting $\rho_{(n)}$ back into Eq. (10) yields the elastic–plastic boundary radius $R_p=R_0/\rho_{(n)}$, and then the radius of each ring in the plastic zone, $r_{(i)}=\rho_{(i)}R_p$, can be calculated.

The displacement of each point can be obtained using the geometric equation:

$$u_r=\varepsilon_\theta r \tag{27}$$

The calculation process described above is based on the GZZ strength criterion. For more detailed information on the reliability of the calculation method and the improvement of the calculation rate, please refer to the author's previous work (Su et al., 2022).

4. Value of key parameter k and sensitivity analysis

The undetermined coefficient k in the calculation of β or GSI_{eq} given by Eq. (7) is related to the input parameters used for the calculation of the excavated tunnel. To investigate the impact of various input parameters on the value of k, the different calculation parameters are selected within a reasonable range and combined randomly to obtain the k value of Eq. (7) for different parameter combinations. Statistical analysis is then performed. The values of the rock mass parameters are provided in Table 1.

Through pre-test analysis, it is found that the excavation radius R_0 and Poisson's ratio ν have a negligible effect on the results (Walton & Diederichs, 2015). Therefore, the constants $R_0=8$ m and $\nu=0.25$ are used in the calculation. Since the smooth blasting design is commonly used in construction and results in minimal disturbance to the surrounding rock, so $D=0$ is taken in calculation (Hoek & Brown, 2019). Due to the random combination of parameters, the obtained displacement may be excessively large or small, which rarely occurs in engineering practice. For such cases, a separate design analysis method is required for large deformation problems, while for the elastic tunnel with small deformation, better stability can be achieved even without support design. Such results are not within the scope of analysis in this paper. Among the 2798 sets of data obtained, 91 % of the predicted k had an R^2 exceeding 0.9, indicating a relatively high overall fitting accuracy.

Fig. 6 shows the results of the statistical analysis performed on the predicted k value according to different input parameter values. It is found that the elastic modulus E has a greater influence on the prediction results, as seen in Fig. 6(a). However, the data is more discrete when $E=1$ GPa, with a relatively large median and data distribution range of the predicted k value, and a low proportion of valid data of 0.054 compared to other cases. The problem with a low proportion of valid data also exits when $p_0=10$ MPa, as shown in Fig. 6(b), so these two cases are not considered in the regression analysis. In practical engineering, the elastic modulus E represents the quality of the rock mass, and the in-situ stress p_0 represents the geomechanical environment of the tunnel. Both of them have a significant influence on the calculation results of tunnel excavation, and are the controlling factors for excavation and design in practical engineering. However, the support stress p_s, which represents construction factors, has little effect on the predicted k value, as shown in Fig. 6(c). The influence of other rock mass parameters on the calculation results is more regular, with the distribution of the predicted k value showing an obvious change law with the variation of the parameters. The data distribution is more uniform for parameters such as σ_c, as shown in Fig. 6(d).

The predicted k values are distributed around -1.8, with data on both sides showing a stepwise distribution. The $(-k)$ follows an approximately lognormal distribution, as shown in Fig. 7(a), which is transformed into a normal distribution by taking the natural logarithm, as shown in Fig. 7(b). Then, multiple linear regression analysis is conducted, with $\ln(-k)$ as the target variable and each input parameter as the independent variable. The prediction model exhibited an R^2 value of 0.961, indicating that the model could effectively describe the impact of each parameter on the value of k. The results of the regression standardized residual are shown in Fig. 8.

The scatterplot of standardized residuals exhibits a V-shaped distribution trend, as shown in Fig. 8(c), suggesting a nonlinear relationship in the regression model. In fact, the relationship between the calculation parameters and the output results should be nonlinear. However, it is often not practical to establish complex nonlinear models to explore the laws between parameters in engineering applications. Instead, simple and convenient linear models are more commonly used within the allowable range of engineering errors. As shown in Fig. 8, the mean of the regression standardized residual is close to 0, the standard deviation is close to 1, the P-P diagram is approximately linear, and the scatter plot is relatively evenly distributed, with 96 % of the standardized residual values falling within the range of -2 to 2. These results indicate that the distribution of standardized residuals follows a normal distribution, and the linear regression model used is reasonable within the allowable error range.

The results of the multiple linear regression model can be used to predict the value of k by input calculation parameters,

$$\ln(-k)=c_0+\sum_{i=1}^{7}(c_iS_i) \tag{28}$$

where S_i represents the input calculation parameter, and c_i represents the corresponding regression coefficient. The value of c_i is shown in Table 2.

In the analysis of the ESS model, the GSI_p can be used as the input parameter first, and the plastic zone radius $R_{p,EPP}$ can be calculated

Table 1
Rock mass parameters.

Parameters	Values				
p_0/MPa	10	20	30	40	50
σ_c/MPa	30	40	50	60	70
E/GPa	1	5	10	15	
ψ/°	5	10	15		
m_i	10	15	20		
p_s/MPa	0	1	2		
χ/×10^3	0.8	1.0	1.2		

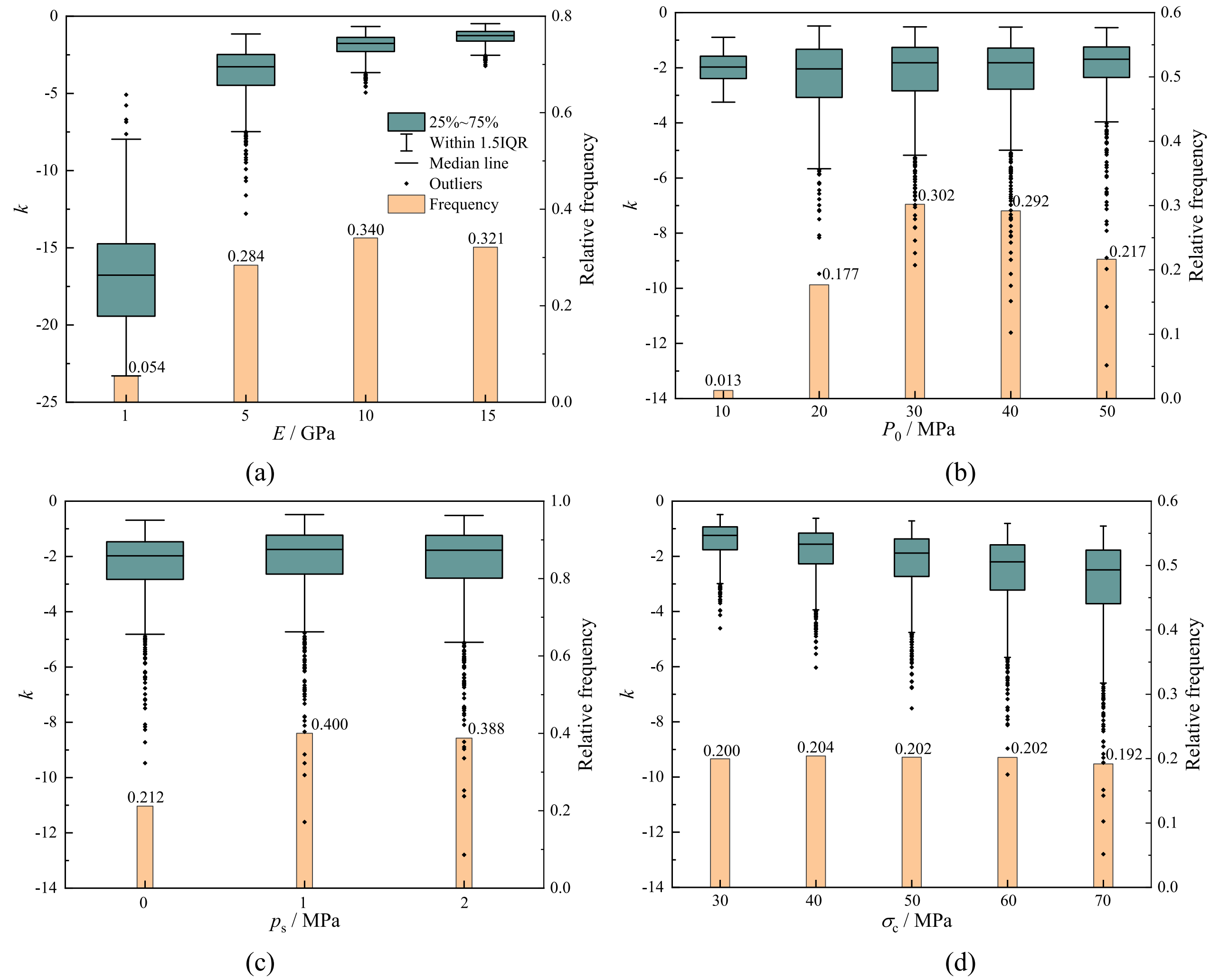

Fig. 6. Boxplots and histograms: (a) E; (b) p_0; (c) p_s; (d) σ_c.

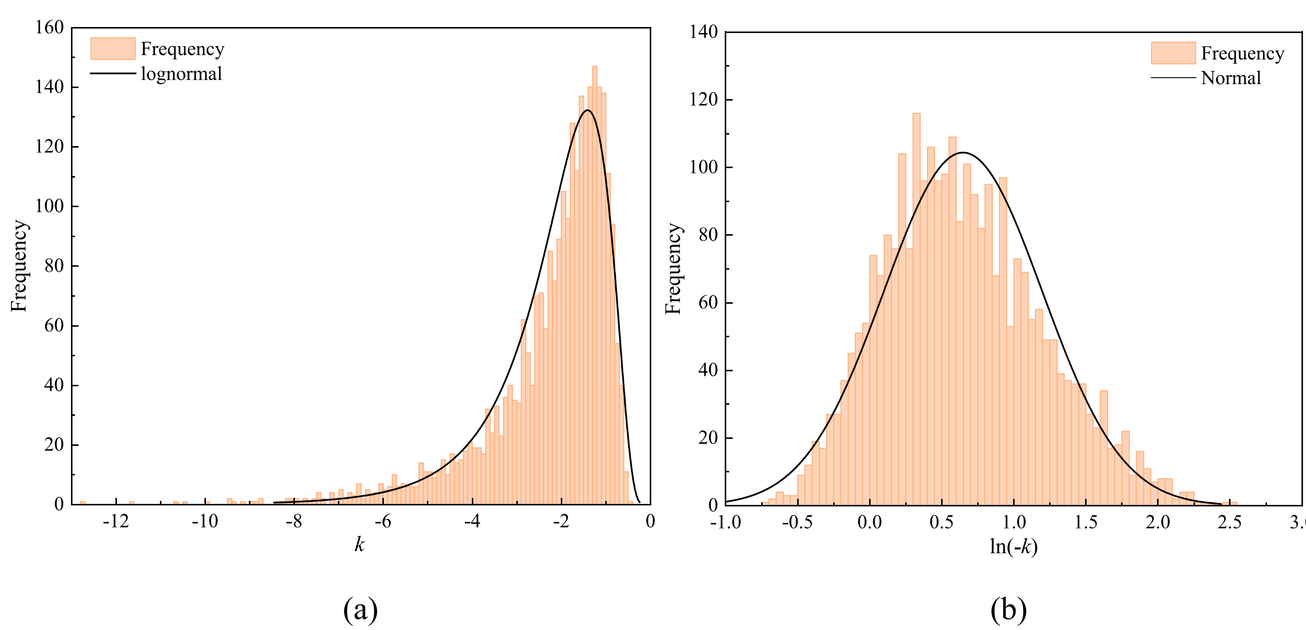

Fig. 7. Distribution of predicted k value: (a) k; (b) $\ln(-k)$.

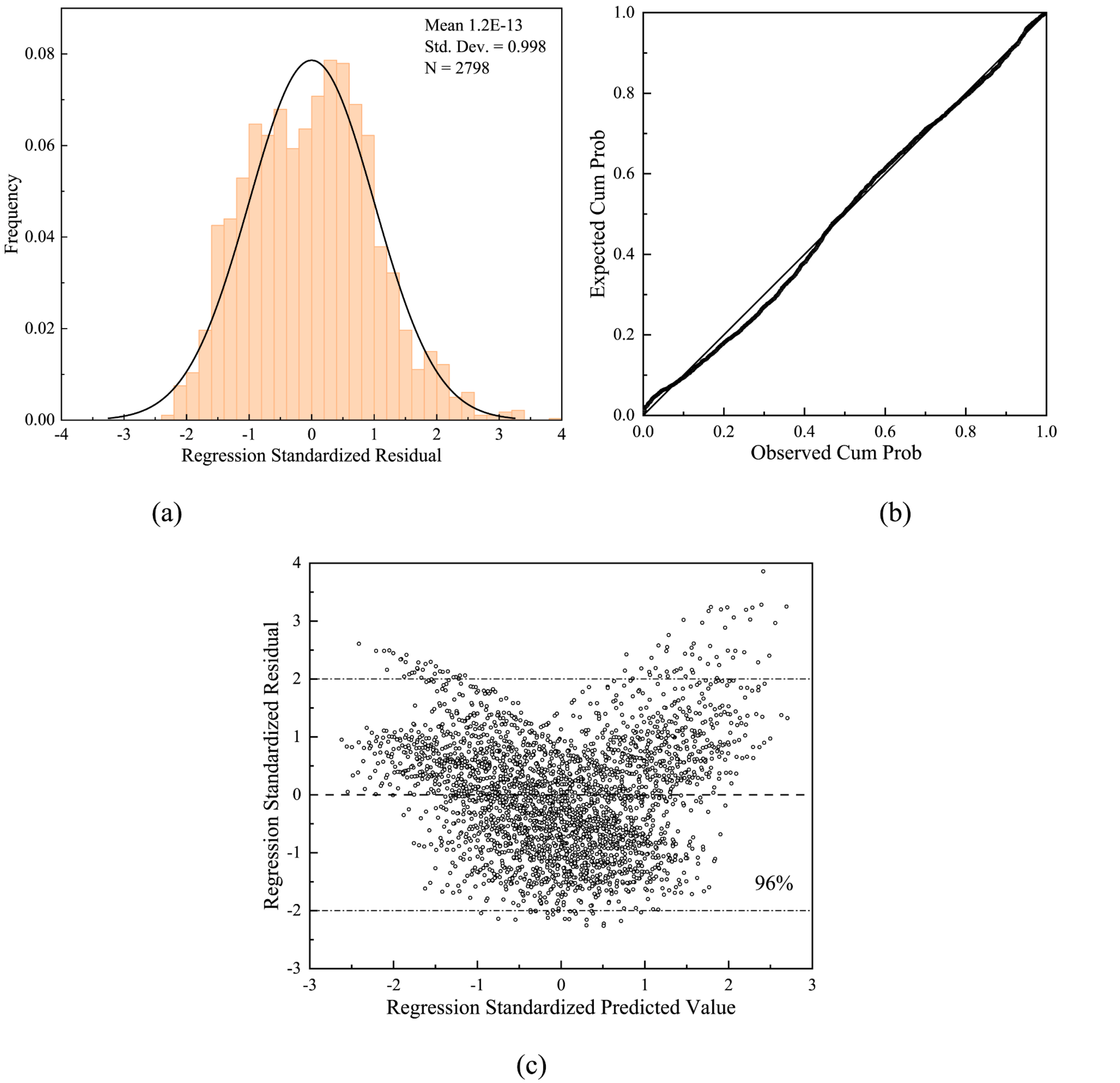

Fig. 8. Regression standardized residual: (a) Histogram; (b) Normal P-P Plot; (c) Scatterplot.

Table 2
Suggested value of c_i (based on extensive case analysis).

$c_i(S_i)$	c_0(Constant)	$c_1(p_0$/MPa)	$c_2(\sigma_c$/MPa)	$c_3(E$/GPa)	$c_4(\psi/^\circ)$	$c_5(m_i)$	$c_6(p_s$/MPa)	$c_7(\chi/\times10^3)$
c_i value	−1.077	0.003	0.016	−0.100	0.022	0.053	−0.072	0.902
Std. Error	0.019	0.000	0.000	0.001	0.001	0.001	0.003	0.013

using the EPP model. Next, the k value can be obtained by Eq. (28). Finally, the equivalent GSI_{eq} for EPP calculation can be obtained by substituting the GSI_p and k into Eq. (7). The calculation process is illustrated in Fig. 9.

5. Verification and discussion

To verify the accuracy and practicality of the proposed method, the following case analyses are conducted.

5.1. Error analysis

The calculation parameters for all the calculation cases in Part 4, along with each GSI value ranging from 20 to 80, $\Delta GSI = 1$, are used to calculate using the calculation process depicted in Fig. 9. The calculation results are compared with those of the ESS model, and the error $T(GSI_{eq})$ generated by the proposed method is recorded, as shown in Fig. 10.

The distribution of $T(GSI_{eq})$ resembles a ladder shape with values on both sides of 0, with an overall small error. Most larger relative errors are typically observed for data points with smaller u_{R0} values, even if the

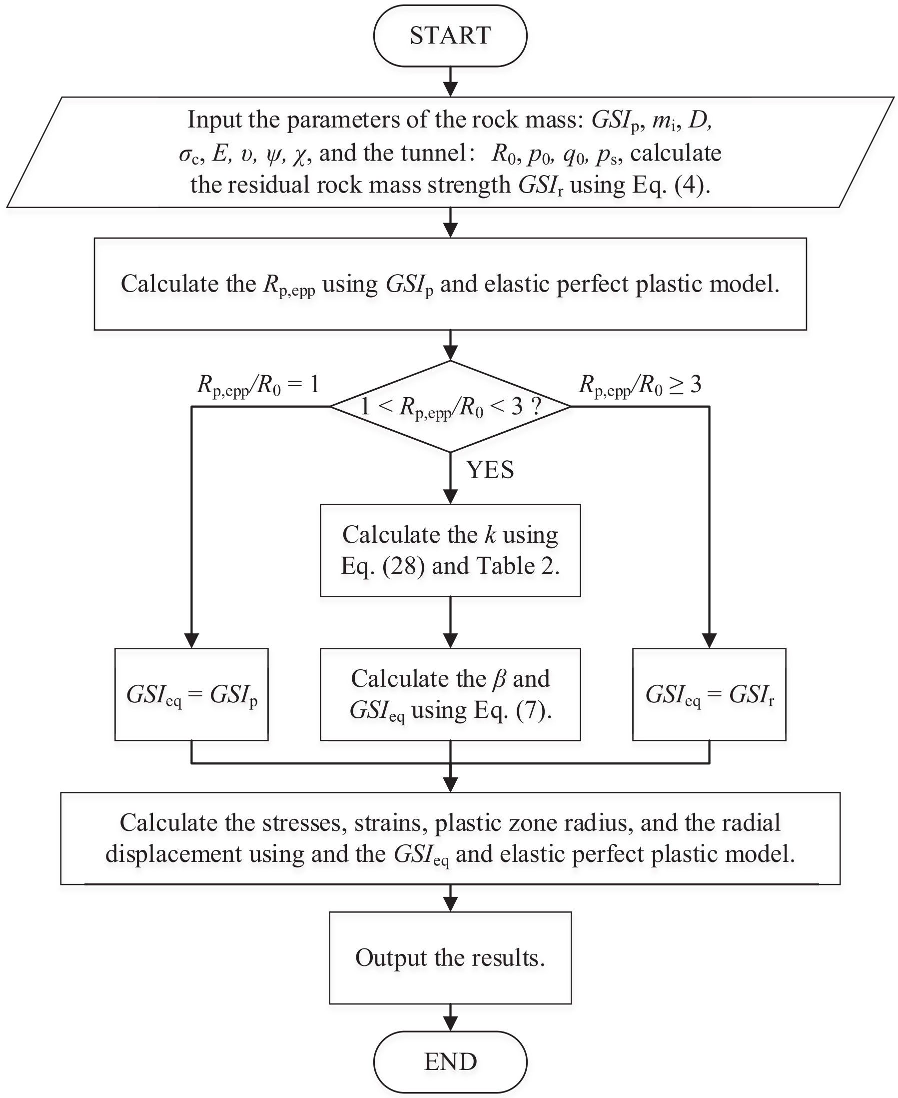

Fig. 9. Flow chart for the calculations.

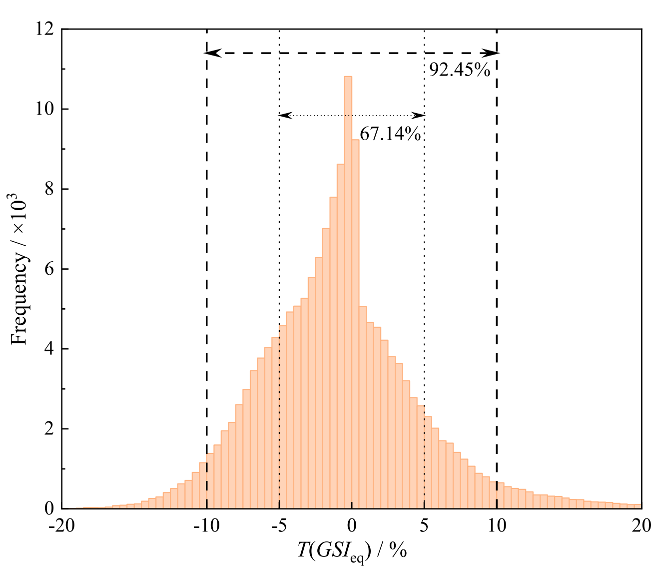

Fig. 10. Error histogram.

absolute error is small. Multiple sources of error exist throughout the calculation process. Although the stress calculation methods used in the two post-peak models differ, the EPP calculation method used in this paper is an improvement within the framework of ESS calculation. The results of the two methods for the same EPP case exhibit almost perfect agreement, with only negligible small errors (Su et al., 2022). The solution for $\ln(-k)$ in Eq. (28) is obtained by extensive parametric analysis and linear regression. Due to the linear model approximately characterizes the nonlinear relationship, errors of varying magnitudes will arise from different parameter combinations. These estimation errors are further amplified during the two exponential operations, namely the conversion of $\ln(-k)$ to k and the calculation of Eq. (7). Thus, the main source of error lies in the selection of the k value. In addition, Eq. (7) gives an approximate exponential function form of GSI_{eq} calculation, which entails some degree of error. Nevertheless, as shown in Fig. 10, the error $T(GSI_{eq})$ for over 90 % of the calculation cases is less than 10 %, and for more than two-thirds cases is even less than 5 %. This demonstrates that the proposed equivalent EPP calculation method and the GSI_{eq} value method, considering the strain-softening characteristics, are reasonable within the allowable engineering error range.

5.2. Stress and displacement curves

To evaluate the accuracy of the proposed method in a particular case, the rock mass parameters in Part 2.2 are used to calculate the stress and radial displacement curves of the ESS and the equivalent EPP models, where $\chi = 1000$. We have obtained $R_{p,EPP} = 12.48$ m ($1.56R_0$) using the calculation process shown in Fig. 9, $k = -2.129$ via Eq. (28), and $GSI_{eq} = 32.98$ using Eq. (7). The calculation results of the two models are shown in Fig. 11.

As the prediction of the GSI_{eq} in this paper is based on the displacement and radial stress at the tunnel wall, the errors of u_r and σ_θ at the tunnel wall between the two models are minimal, while some error is observed in the tangential stress. Due to the *GSI* of the EPP model is constant in the plastic zone, the slope of the σ_θ curve in the plastic zone has no significant variation, whereas the curve of the ESS model can effectively distinguish between the softening zone and the plastic flow zone by the curve slope. The σ_θ at the plastic radius calculated using the proposed method is relatively small for $GSI_{eq} \leq GSI_p$. Conversely, the σ_θ at the tunnel wall is larger for $GSI_{eq} \geq GSI_r$, resulting in a large error in the σ_θ curves of the two models. The stress and displacement curves of the two models in the elastic zone are consistent. The σ_θ at the plastic radius calculated by GSI_{eq} is relatively small. For the case of the same σ_θ stress curve in the elastic zone, a smaller σ_θ at the plastic radius implies that it will enter the plastic zone earlier, resulting in a wider plastic zone and a larger R_p, as shown in Fig. 11. Because the R_p calculated by the proposed method is too large, in the calculation and analysis considering the range of the plastic zone, such as LDP, directly using the plastic radius of the proposed model may lead to large errors. The radial stress σ_r in the plastic zone calculated by the proposed method will be larger, but the overall deviation of the radial stress curve between the two models is small. Although the error in the stress distribution will cause inevitable deviations in the displacement curve, but the overall distribution of the displacement curve of the two models remains consistent. Despite the deviation of the σ_θ curve calculated by the proposed method may be relatively large, the distribution of radial displacement and stress is more concerned in engineering, and the distribution curve of u_r and σ_r calculated in this paper is in good agreement with the ESS method.

5.3. GRCs

The calculation parameters in 5.2 are adopted to calculate the displacement of the tunnel wall under various support stresses, and the resulting GRCs are illustrated in Fig. 12.

The GRCs calculated by the two models exhibit small discrepancies,

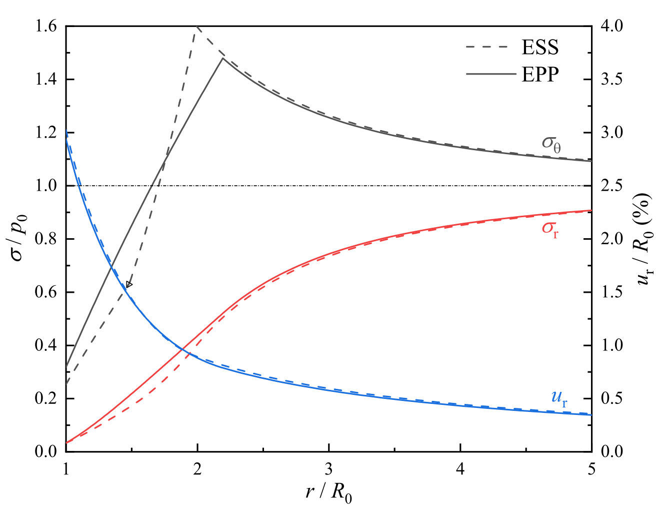

Fig. 11. Comparison of stress and radial displacement results of two models.

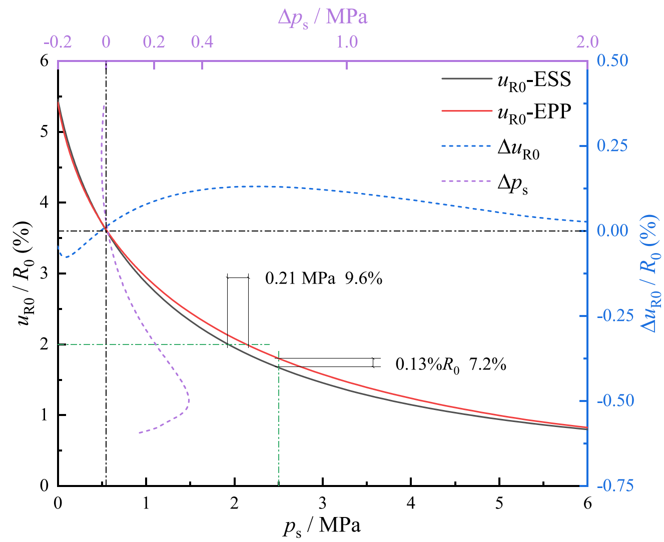

Fig. 12. Comparison of GRCs.

as shown by the black and red solid lines in Fig. 12. For example, when $p_s = 2.5$ MPa, the difference in u_{R0} calculated by the two models is only 0.13 $\%R_0$ (7.2 %). The error tends to be higher in the middle but lower at both ends of different p_s, and the overall error is small, ranging between -0.08 $\%R_0$ and 0.13 $\%R_0$, as shown by the blue dotted line in Fig. 12. Similarly, the error in p_s lies between -0.02 MPa and 0.35 MPa, as shown by the purple dotted line in Fig. 12. When the tunnel control deformation is 2 $\%R_0$, for instance, the difference in p_s calculated by the two models is only 0.12 MPa (9.6 %), falling within an acceptable range of error.

During the calculation of the GRC, different support stresses correspond to different $R_{p,EPP}$ and k values. As a result, each support stress value (p_s) in the proposed equivalent EPP method requires two iterative solution processes, namely the calculation of the boundary conditions using GSI_p and GSI_{eq} as parameters, respectively. In contrast, the number of iterations for the ESS method is $n + 1$ for each support stress. If the number of p_s solutions is m, the corresponding number of iterations for the EPP and ESS methods are 2 m and $m(n + 1)$, respectively. As the number n of plastic zone divisions increases exponentially, the calculation time of the EPP method increases little, while that of the ESS method increases exponentially, as shown in Table 3. Most of the running time for the calculation program is consumed in the iteration process, and the iteration time of the ESS method always accounts for more than 99 % of the total time. As the number of iterations increases, the calculation time also increases. However, the iteration time of the EPP method remains almost unchanged, and the increase in the total program time is caused by the increase in other common calculation times. Therefore, the proportion of iteration time decreases with the increase of n. The finite difference method can achieve higher calculation accuracy as n increases, and the EPP method offers significant advantages in terms of calculation speed and even better calculation accuracy (Su et al., 2022). It takes little time to calculate a single case for both methods. For complex problems, such as GRC, the required m and n values become enormous, the method proposed in this paper can save a significant amount of time.

5.4. Effect of p_0

The geomechanical conditions of the surrounding rock play a crucial role in the stress distribution after excavation. The rock mass calculation parameters in Part 5.2 and $p_0 = 20$, 30, 40, and 50 MPa are adopted to analyze the surrounding rock response under varying initial in-situ stress conditions.

Table 3
Calculation time ($m = 100$).

n	Total time/s		Number of iterations		Iteration time/s		Iteration time ratio/%		Efficiency ratio
	EPP	ESS	EPP/2 m	ESS/$m(n+1)$	EPP	ESS	EPP	ESS	
10	1.493	7.845	200	1100	1.456	7.783	97.52	99.21	5.3
100	1.512	67.920	200	10,100	1.440	67.546	95.24	99.45	44.9
1000	1.812	636.256	200	100,100	1.439	632.701	79.42	99.44	351.1
10,000	4.248	6429.785	200	1,000,100	1.233	6397.443	29.03	99.50	1513.6

As shown in Fig. 13(a), as the increase of p_0, the stress in the plastic zone of the surrounding rock increases, and the range of the plastic zone expands. Since the GSI_r of the same rock mass remains constant, the stresses in the flow zone, calculated from the same GSI_r value, lie on the same line, as shown by the purple line in Fig. 13(a). The range of the flow zone varies for different p_0 conditions, while the stress in the flow zone is the same. The distribution law of σ_r is similar to that of σ_θ. As p_0 increases, the range of the plastic zone and the proportion of the flow zone increase. Consequently, the predicted GSI_{eq} value gradually decreases and approaches to GSI_r, and the σ_θ curve gradually converges towards the radial stress trend line of the flow zone. As the buried depth (p_0) increases, the stress curve in the plastic zone of the surrounding rock approaches to the EPP state, and it is more reasonable to use the EPP model for calculation.

As p_0 increases, the range of the plastic zone increases approximately linearly, as shown in Fig. 13(a). However, the displacement of the tunnel wall shows an exponential increase, as shown in Fig. 13(b). The deformation capacity of the surrounding rock in the plastic zone is strong, so the excavation-induced deformation is mainly plastic deformation under high ground stress conditions. Thus, the proportion of plastic deformation to the displacement of the tunnel wall exhibits a non-linear increase trend with p_0. When $p_0 = 50$ MPa, nearly 90 % of the displacement results from the deformation in the plastic zone, as shown in Fig. 13(b). The plastic radius and plastic deformation gradually increase with the buried depth (p_0). The excavation stability of the deep tunnel is primarily controlled by the plastic stress and plastic deformation of the surrounding rock.

The difference of the displacement curves between the two models increases with the increase of p_0, as shown in Fig. 13(b). Nonetheless, the relative error can still be limited within 4 %, while the absolute error increases alongside the displacement value itself.

5.5. Effect of GSI

The rock mass calculation parameters outlined in Part 5.2 are adopted to calculate the results for GSI values ranging from 20 to 80 using the proposed method. The results are compared with those of the ESS method. With $\Delta GSI = 1$ and predicted $k = -2.129$, the stress and displacement curves are calculated separately for GSI values of 20, 30, 40, 60, and 80, as shown in Fig. 14.

The proposed equivalent EPP method and the ESS method provide relatively similar calculation results with small errors ranging from −4.0 % to 6.4 %, as shown in Fig. 14(a). The tangential stress and radial displacement curves also exhibit small errors under different GSI conditions, as shown in Fig. 14(b) and (c). When $GSI \le 22$, $GSI_{eq} = GSI_r$ is directly taken for calculation in the proposed method for $R_{p,EPP}/R_0 > 3.0$. Due to the different value methods of GSI_{eq}, there is a jump in the error curve between GSI equal to 22 and 23, as shown in Fig. 14(a). When $GSI = 80$, $R_{p,EPP}/R_0$ is close to 1, and the error is close to 0. The error fluctuates around 0 and gradually decreases to 0 when GSI is larger or smaller, as shown in Fig. 14(a), indicating that it is reasonable to directly take the peak or residual value as $R_{p,EPP}/R_0$ is equal to 1 or greater than 3, respectively.

As the GSI decreases, the surrounding rock becomes more fragmented, leading to a decrease in the bearing capacity and the plastic zone stress. In addition, the plastic zone becomes more widely distributed, and the proportion of the flow zone to the plastic zone increases, as shown in Fig. 14(b). Fig. 14(c) presents a statistical diagram of the proportion of flow zone displacement to the entire plastic displacement under different GSI conditions. The quality of rock mass in the flow zone is lower than that in the softening and elastic zones, resulting in a stronger ability to generate displacement. Therefore, the residual flow zone has a great influence on the plastic deformation of the surrounding

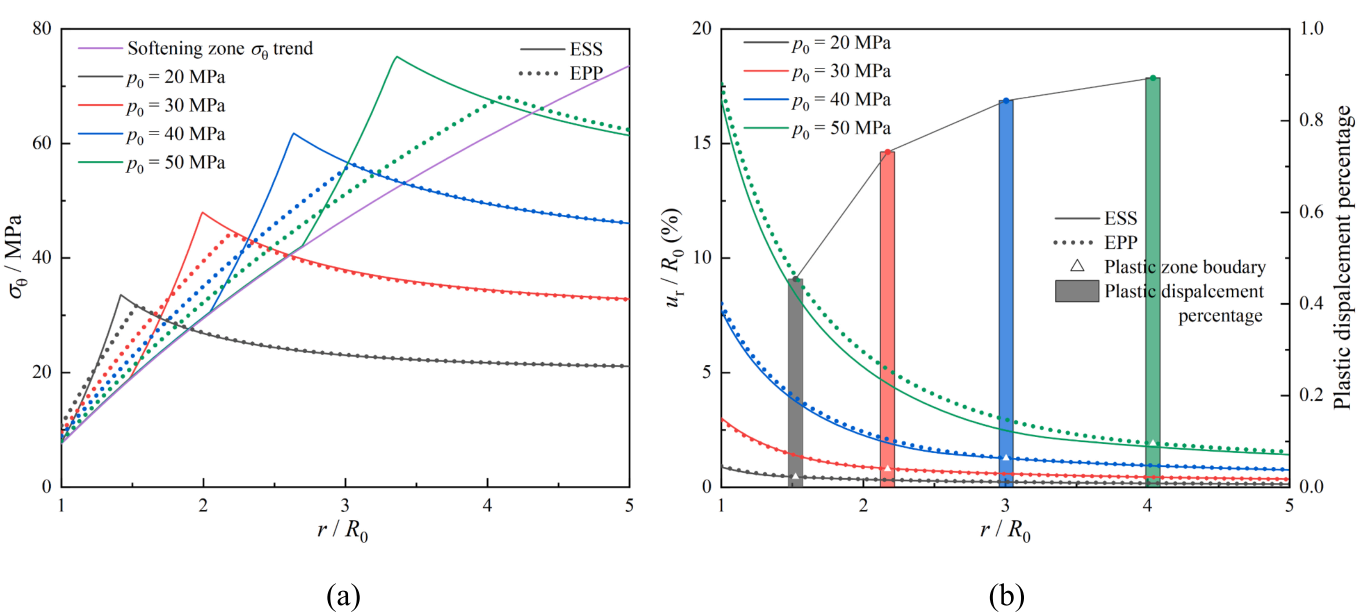

Fig. 13. Comparison of different p_0 conditions: (a) σ_θ; (b) u_r.

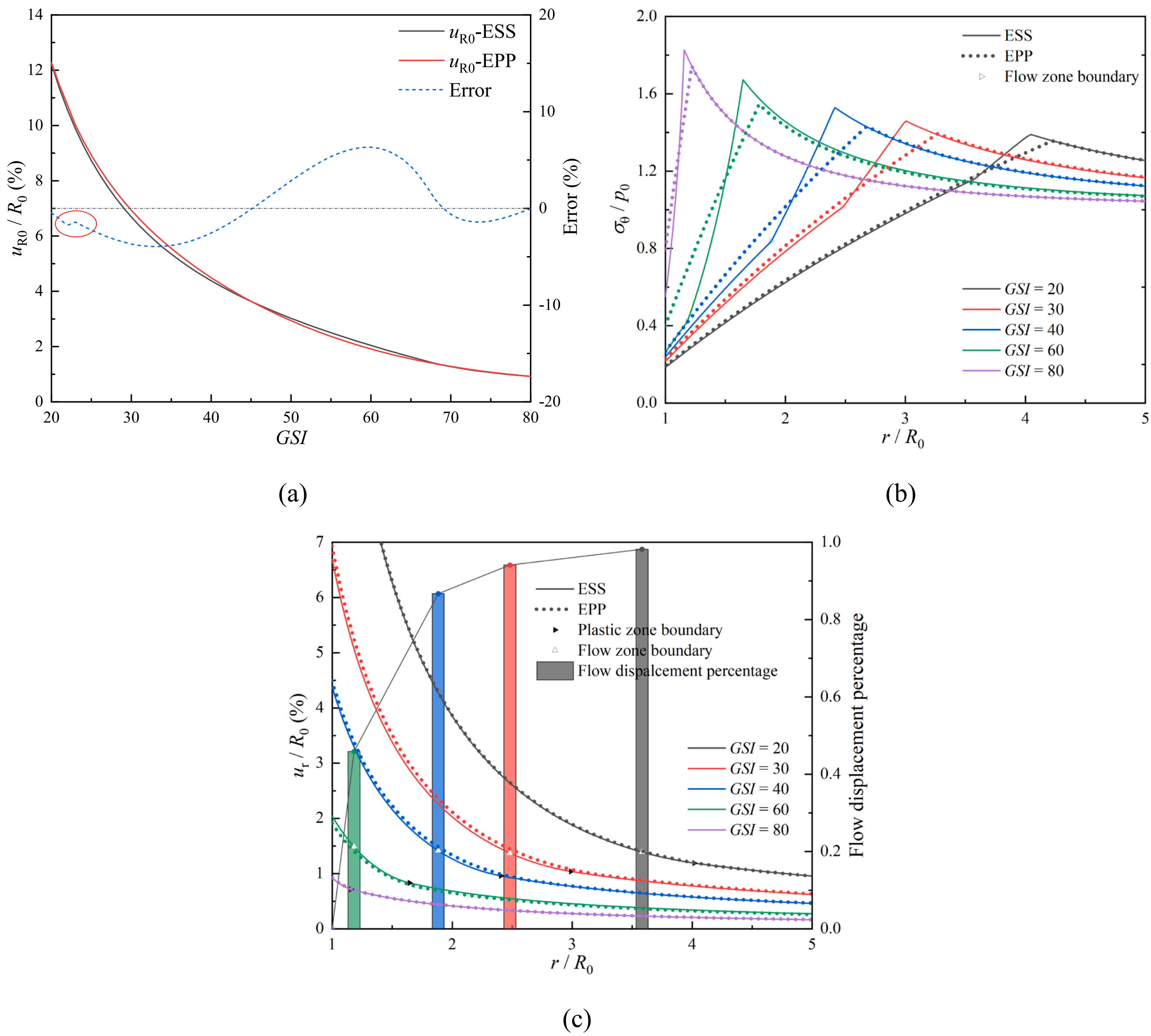

Fig. 14. Comparison of different *GSI* conditions: (a) u_{R0}; (b) σ_θ; (c)u_r.

rock (Cui et al., 2019). When $GSI = 80$, the plastic zone is small, and there is no flow zone; when $GSI = 60$, 46 % of the plastic deformation occurs in the flow zone; and when $GSI = 20$, the plastic zone is widely distributed, with more than 98 % of the plastic deformation occurring in the flow zone, as shown in Fig. 14(c).

When $GSI = 20$, the GSI_p is close to GSI_r, and the curves of the softening zone and the flow zone are not significantly different. Therefore, the residual strength value can be directly adopted in the calculation of the EPP model by ignoring little strength in softening area, resulting in a good match between the σ_θ curves of the two models, as shown by the black dotted line in Fig. 14(b). This is consistent with Hoek and Brown's recommendation to use the EPP model for weak surrounding rock with $GSI < 25$ (Hoek & Brown, 1997). They also pointed out that the EBP model should be used for relatively complete surrounding rock with $GSI > 75$. In the case of highly qualified rock mass with $GSI = 80$, the stress in the plastic zone decreases rapidly and the curve type is similar to that of the EBP model, as shown by the purple line in Fig. 14(b). As the *GSI* increases further, the plastic zone continues to shrink until it disappears. At this point, $R_{p,EPP}/R_0 = 1.0$, and the surrounding rock only has an elastic zone, resulting in identical outcomes of the EPP and EBP models.

6. Conclusion

By determining the equivalent *GSI* strength parameters, the elastic perfect plastic model can be utilized to effectively analyze the strain-softening mechanical response of the surrounding rock. The proposed method has distinct advantages in computational efficiency when solving the GRC and analyzing the stability of surrounding rock. Furthermore, it allows for the convenient consideration of the influence of three-dimensional rock mass strength, such as the GZZ strength criterion, on the mechanical response of surrounding rock. Although the calculation results of σ_θ based on the EPP model with GSI_{eq} differ from those of the ESS model, the σ_r and u_r curves of the two models agree well, and the error of the GRC is minimal. The statistical analysis shows that over 90 % of the errors $T(GSI_{eq})$ fall within 10 %, indicating the rationality and reliability of the proposed method.

After comprehensively considering the influence of in-situ stress, rock mass characteristics, and engineering factors, an extensive statistical analysis of GSI_{eq} is conducted. The results reveal that the rock mass elastic modulus E and initial ground stress level p_0 significantly affect GSI_{eq}, while the support reaction force p_s contributes little to that. The key parameter $(-k)$ in GSI_{eq} calculation confirms a lognormal distribution, and the regression model for $\ln(-k)$ has a high accuracy ($R^2 = 0.961$), indicating the reliability of the given method of k value. The value of k can be directly selected according to Table 2, enabling a forward analysis of GSI_{eq}.

The case analyses show that with the increase of the in-situ stress level p_0 (buried depth), GSI_{eq} gradually decreases and approaches to GSI_r, and the stress curves in the plastic zone of the surrounding rock of the two models become more similar. The proportion of plastic deformation to the total deformation increases nonlinearly with buried depth, implying that the stability of the deep tunnel is primarily controlled by plastic stress and deformation. It is worth mentioning that when $R_{p,EPP}/R_0 > 3.0$, usually $GSI < 25$, the deformation of surrounding rock is primarily plastic deformation in the residual flow zone. Therefore, a small amount of strength in softening zone can be ignored, and $GSI_{eq} = GSI_r$ can be directly used as the calculate input parameters for the EPP model. For highly qualified rock mass, such as $GSI > 75$, $GSI_{eq} = GSI_p$ can be directly used for EPP calculation.

Although the methods proposed in this paper are limited in scope, especially in analyzes that consider the range of the plastic zone, it appears to provide a novel idea to simplify the analysis of strain-softening rock masses.

CRediT authorship contribution statement

Chenlong Su: Investigation, Methodology, Software, Data curation, Formal analysis, Writing – original draft. **Hehua Zhu:** Supervision, Conceptualization, Project administration, Resources. **Wuqiang Cai:** Investigation, Methodology, Formal analysis, Project administration, Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing. **Wei Wu:** Methodology. **Qi Zhang:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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